Course Information

- **ECE 20869:** Discrete Mathematics for Computer Engineering
- **CRN:** 36176
- **Meeting time:** TuTh 10:30-11:45 (MTHW 210)
- **Instructional Modality:** Face-to-Face
- **Course credit hours:** 3
- **Prerequisites:** ECE 20875 or ECE 26400 [may be taken concurrently]

Course Description

This course introduces discrete mathematical structures, with a focus on developing problem-solving skills and abstract reasoning. Students will attain substantial experience formulating and reasoning about a wide range of discrete mathematics problems. Topics generally include sets, sequences, relations, mappings, recursive definition, formal logic syntax and semantics, mathematical induction, recursion and loop invariants, counting and combinatorics, countability, asymptotic complexity, finite-state automata, regular expressions, and non-determinism.

Learning Resources, Technology & Texts

Course Websites

- **Brightspace** (lecture videos and notes, weekly practice problems, grades): https://purdue.brightspace.com/d2l/home/1489070

Required Textbook

- Mathematical Structures for Computer Science, 7th Edition, Judith L. Gersting, W. H. Freeman, 2014, ISBN No. 978-1429215107

Learning Outcomes

A student who successfully fulfills the course requirements will have demonstrated:

i. an ability to define, reason about, and operate on sets using basic set operations and recursive definition. [1,7]

ii. an ability to represent and verify entailment arguments in predicate logic, both syntactically and semantically. [1,3,7]

iii. an ability to argue carefully using a variety of informal proof techniques, including mathematical induction. [1,3,7]

iv. an ability to count in a wide range of settings including permutations, combinations, countable infinities, and recurrence equations. [1,7]

v. an ability to define and reason about functions and binary relations and work with their useful properties. [1]

vi. an ability to define finite-state automata and regular expressions for languages, and relate these formalisms using non-determinism. [1,7]

vii. an ability to formulate algorithmic problems as decision problems and reason about their decidability and scalable solvability. [1,7]

Assignments and Exams

Exams

We will have evening midterms and final. The exams are closed book and closed notes. The tentative times and topics for the exams are below:

Exam	Time	Location	Topics (tentative)
Midterm 1	Thursday, February 12th, 8:00-10:00pm	PHYS 112 PHYS 203	Propositional logic, relations, functions predicate logic, mathematical proofs
Midterm 2	Tuesday, March 31st, 8:00-10:00pm	FRNY G140 MSEE B012	Induction, program proofs, recurrences, sets and combinatorics
Final	TBD	TBD	Algebraic structures, finite state machines, Turing machines

If you foresee an unavoidable conflict with a scheduled exam, you must notify the instructor via email as early as possible, but no later than **one week prior to the exam date**, to discuss alternative arrangements. For last-minute emergencies or illnesses that arise after this deadline, requests for accommodations will be evaluated on a case-by-case basis and will require formal supporting documentation, such as a signed physician's note or an official university absence letter.

Due to the extra hours for midterms, the meetings on Tuesday, February 17th, Thursday, April 2nd, and Thursday, April 30th are (tentatively) to be omitted.

Practice Problems

There will be weekly exercises for your practice and test preparation. The exercises will not be collected or graded. In lieu of traditional homework submission, we will hold active learning sessions in class to discuss these practice problems. Solutions to each exercise will be posted after the classroom discussion.

Grading Scale

Grades will be assigned as follows:

90% — Tests (2 midterms and 1 final, 30% each)
10% — Classroom participation

You may earn classroom participation credit in many ways, e.g., presenting a solution in an active learning session, asking a good question (in class or on Piazza), reporting a mistake in an assignments or exam, etc.

There may be a constant curve (i.e., all grades will be increased by a fixed amount) for individual exams at the instructor's discretion. Your course grade will be determined using an absolute scale: 97–100: A+; 91–97: A; 88–91: A-; and continuing down.

Course Schedule (tentative)

Week	Topic
1	*Propositional logic*
2	*Sets, relations, functions*

3	*Predicate logic*
4	*Mathematical proofs*
5	*Slack/Midterm 1*
6	*Induction*
7	*Program proofs*
8	*Program proofs (continued)*
9	*Recurrences*
10	*Spring break*
11	*Combinatorics*
12	*Slack/Midterm2*
13	*Algebraic structures*
14	*Finite state machines*
15	*Turing machines*
16	*Slack/Review*

* Schedule and tests subject to change. Lecture slides will be posted on the course website.

Academic Integrity

We hope (and assume) that you are all honest and have no intentions of cheating. Cheating ruins the experience for everyone and we will pursue appropriate penalties if we catch someone cheating. Please don't -- it's not worth it, both for you and for us. You can find Purdue's student guide for academic integrity at https://www.purdue.edu/odos/academic-integrity/ .

The use of generative artificial intelligence (AI) tools (e.g., ChatGPT, Gemini) on any assessed work (e.g., exams) is strictly prohibited. Using an AI tool will be treated as an instance of academic dishonesty.

If we do catch someone cheating for the first time, the following penalties will be applied: zero on the quiz, and we will subtract 10% from your final average in the course or reduce your letter grade by one category (e.g., B to B-), whichever results in the lower grade.

For second cheating offense of any kind, you will fail this course.

Nondiscrimination Statement

A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in the Academic Resources table on your Brightspace homepage.

Accessibility

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at: **drc@purdue.edu** or by phone: 765-494-1247, as soon as possible.
If the Disability Resource Center (DRC) has determined reasonable accommodations that you would like to utilize in my class, you must release your Course Accommodation Letter to me. Instructions on sharing your Course Accommodation Letter can be found by visiting: **https://www.purdue.edu/drc/students/course-accommodation-letter.php** Additionally, you are strongly encouraged to contact me as soon as possible to discuss implementation of your accommodations.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.